

Logical operators -

] and :-

Condition 1 and Condition 2 -

if both Condition 1 & Condition 2  
are true then it returns  
true.

2] or

Condition 1 or Condition 2 —

if any one condition is true,  
it returns true.

eg.

age = 22

nationality = "Indian"

Condition

True

False

True False

True

if (age >= 18) and (nationality == "Indian"):

→ print("you are eligible for voting")

else:

print("you are not eligible for voting")

eg.

is\_weekend = True

is\_holiday = False

True  
↓

if is\_weekend or is\_holiday:

→ print("you can relax")

else:

print("you need to work")

and

Condition 1

True

True

False

False

Condition 2

True

False

True

False

— True

— False

— False

— False

OR

condition 1

True

True

False

False

condition 2

True

False

True

False

True

True

True

False

Conditional expression :-

- This is the compact way of writing if-else statements.
- we can write if-else statements in one line.



1) First it evaluates conditional expression, if condition is true, it prints  $exp1$ .

← if condition is False  
it prints  $exp2$ .

Q.

```
1 age = 22
```

```
2 if (age >= 18):
```

```
3     print("you are eligible for  
    voting")
```

```
4 else:
```

```
5     print("you are not eligible  
    for voting")
```

all() and any()  
↓  
and()

or()

all() — all built in function  
returns true if all  
conditions are true.

ex.

$a = 1$ ,  $b = 2$ ,  $c = 3$

$\text{result} = \text{all} \left( \overset{1 < 5 \checkmark}{a < 5}, \overset{2 < 5 \checkmark}{b < 5}, \overset{3 < 5 \checkmark}{c < 5} \right)$

$\text{print}(\text{result}) \longrightarrow \underline{\text{True}}$

if  $\text{result}$  :  
 $\text{print}(\text{"a, b \< are less than 5"})$

else:  
 $\text{print}(\text{"not less than 5"})$

any() - any built in function  
returns true, if any one  
condition is true

$a = 10$  ;  $b = 20$  ;  $c = 300$

print (any ( $a < 5$  ,  $b < 5$  ,  $c > 100$ ))

$\begin{matrix} 10 < 5 \\ \times \end{matrix}$  ,  $\begin{matrix} 20 < 5 \\ \times \end{matrix}$  ,  $\begin{matrix} 300 > 100 \\ \checkmark \end{matrix}$

$\Rightarrow$  True

~~for~~

